PROBLEM-BASED ASSIGNMENT

Development of a Heuristic Function for Parking Space Allocation

Artificial Intelligence

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1. Introduction

Artificial Intelligence can be used to solve problems where there may be many possible solutions. Parking space allocation is one such problem. A college campus can contain several parking areas for two-wheelers, cars, buses and staff vehicles. Students, staff and visitors arrive at different times and travel to different blocks of the campus.

A parking system should not only look for an empty space. An empty space may be far from the destination, may be meant for another type of vehicle, may be reserved for staff or may be located in a lane that is already crowded. Therefore, the system needs a way to estimate how suitable each parking space is for a vehicle. A heuristic function provides this estimate.

2. Problem Statement

The problem is to develop a heuristic function for selecting a suitable parking space for a vehicle inside a college campus. The system should evaluate each possible space using factors such as walking distance to the destination, mismatch of space size, congestion near the space and parking restrictions.

Instead of checking every possible combination of vehicles and spaces, the heuristic function assigns a score to each candidate allocation. The search process can then use the score to prefer allocations that appear more suitable. Since the objective is to minimize the cost of the allocation, a lower heuristic value represents a better allocation.

3. Need for the Proposed System

- Large campuses may have several parking areas that a vehicle can use.

- The nearest empty space is not always the most suitable space.
- Drivers may waste time searching for a space, which increases traffic inside the campus.
- Using a large bay for a small vehicle wastes valuable space.
- Crowded lanes and entrances can delay vehicles and cause blockages.
- A heuristic provides a simple way to compare alternative parking spaces.

4. Scope

The proposed approach can be used for allocating parking spaces in a college campus, hostel area or similar shared parking facility. It can be extended later by adding information such as live occupancy sensors, event-day parking rules, electric vehicle charging spaces and accessibility requirements when such information is available.

5. Objectives

The main objectives of the parking space allocation heuristic are:

- To find a suitable parking space for a given vehicle.
- To reduce the walking distance from the space to the destination.
- To match the vehicle type with the size of the space.
- To avoid crowded lanes and entrances whenever possible.
- To avoid spaces that are reserved or restricted for other users.
- To provide a good allocation without checking every possible allocation.

6. Input

The system can use the following information:

- Vehicle details, such as vehicle type and arrival time.
- Destination of the driver, such as Academic Block.
- Type of user, such as student, staff or visitor.
- Location and type of each parking space, such as two-wheeler, car or large bay.
- Availability of each parking space.
- Walking distance from each space to the campus blocks.
- Congestion level of the lane or entrance near each space.
- Restrictions on each space, such as staff-only or reserved.

7. Representation of the Allocation Problem

The problem can be represented as a graph with two groups of nodes. One group contains the vehicles that need parking and the other group contains the parking spaces. A line between a vehicle and a space means that the space is being considered for that vehicle. Each space stores information such as type, zone, distance, congestion and restrictions.

```
Vehicle ---- Slot P1
   |
   |-------- Slot P2
   |
   |-------- Slot P3
```

An example of parking space information is shown below for a driver going to the Academic Block.

Slot	Slot Type	Zone	Distance to Academic Block	Status
Slot P1	Car	North Lot	150 m	Free
Slot P2	Large bay	Main Lot	80 m	Free
Slot P3	Car	Staff Lot	40 m	Free (staff only)

For example, a student arriving by car may be allocated to Slot P1, Slot P2 or Slot P3. The heuristic function evaluates each candidate allocation rather than relying on only one measurement.

8. Assumptions

- The parking spaces and their locations, types and restrictions are known to the system.
- Walking distance is available or estimated for each space and destination.
- Size mismatch, congestion and restrictions can be represented using numerical penalty values.
- The allocation is considered a minimization problem.
- The weights used in the heuristic can be adjusted according to the campus requirement.

9. Heuristic Function

A heuristic function assigns a numerical score to a candidate allocation. The score combines the factors that are important for parking space allocation. The function used in this assignment is:

$$H(A) = W1(D) + W2(S) + W3(Cg) + W4(R)$$

Where:

- A = Candidate allocation (a vehicle assigned to a parking space).
- D = Walking distance from the parking space to the destination, in metres.
- S = Size mismatch penalty (0 if the space type matches the vehicle, 1 if a larger space is used, 2 if a much larger space is used).
- Cg = Congestion penalty of the lane or entrance near the space (0 = free, 1 = light, 2 = moderate, 3 = heavy).
- R = Restriction penalty (0 if the driver is allowed to use the space, 1 if the space is reserved or restricted).
- W1, W2, W3 and W4 = Weights assigned to the four factors.

For the example in this assignment, the weights are selected as follows:

$$H(A) = 1(D) + 30(S) + 20(Cg) + 100(R)$$

When many vehicles are allocated together, the heuristic value of the complete allocation is the sum of the values of all vehicle-space pairs.

Explanation of the Terms

Walking distance is included because a nearer space saves the driver time and effort. Size mismatch is included because using a large bay for a small vehicle wastes space that another vehicle may need. Congestion is treated as a penalty because a crowded lane can delay the driver and other vehicles. Restriction is given the largest weight because reserved or staff-only spaces should almost never be given to other users.

The weights determine the relative importance of the factors. In this example, restriction, size mismatch and congestion receive larger penalty weights so that a space is not selected only because it is close to the destination.

10. Why a Heuristic is Useful

- It reduces the amount of search needed when many spaces and vehicles are involved.
- It combines several practical factors into one score.
- It allows the system to compare spaces using the same measurement.
- It can be used with search methods such as Hill Climbing.
- It can be modified when the requirements of the campus change.

11. Interpretation

Because this is a minimization problem, a lower value of H(A) indicates a more suitable allocation according to the selected criteria. The heuristic does not claim that the allocation is mathematically perfect; it gives a practical estimate that helps the search process make a better choice.

12. Worked Example

Suppose a student arrives by car and wants to go to the Academic Block. Two spaces are free and open to students, Slot P1 and Slot P2. Slot P3 is staff-only and is not considered. The estimated values of the two spaces are shown below.

Slot	Walking Distance D (m)	Size Mismatch S	Congestion Cg	Restriction R	Heuristic
Slot P1	150	0	1	0	170
Slot P2	80	1	2	0	150

Slot P1 is a normal car space in the North Lot. Slot P2 is a large bay in the Main Lot, so using it for a car gives a size mismatch penalty of 1.

Slot P1 Calculation

H(P1) = 1(150) + 30(0) + 20(1) + 100(0)

H(P1) = 150 + 0 + 20 + 0

H(P1) = 170

Slot P2 Calculation

H(P2) = 1(80) + 30(1) + 20(2) + 100(0)

H(P2) = 80 + 30 + 40 + 0

H(P2) = 150

The calculations show that Slot P2 has the lower heuristic value. Although Slot P2 has a size mismatch and a higher congestion penalty, its shorter walking distance results in the lower total score.

13. Comparison of the Parking Spaces

- Slot P1 matches the vehicle type and has no size mismatch penalty.
- Slot P1 is farther from the Academic Block and has lighter congestion.
- Slot P2 is closer to the Academic Block.
- Slot P2 has a size mismatch penalty and higher congestion.

- After combining all four factors, Slot P2 receives the lower heuristic value.
- Therefore, the search method would prefer Slot P2 for this example.

The example demonstrates why a heuristic can be more useful than considering only one factor. It also shows the importance of the weights: if large bays must be kept free for buses, the weight of S should be increased. In an actual implementation, the values and weights should be chosen using appropriate campus data.

14. Hill Climbing Approach

A simple Hill Climbing method can be used with the heuristic function. Hill Climbing starts with an initial allocation and repeatedly looks for a neighboring allocation that has a better heuristic value. A neighboring allocation is created by moving one vehicle to a different parking space or by exchanging the spaces of two vehicles.

15. Algorithm

1. Start with an initial allocation of all vehicles to parking spaces.
2. Calculate the heuristic value of the current allocation.
3. Generate a neighboring allocation by changing the space of one vehicle.
4. Calculate the heuristic value of the neighboring allocation.
5. Compare the new value with the current value.
6. If the neighboring allocation has a lower value, accept it as the current allocation.
7. Continue until no better neighboring allocation is found.
8. Return the final allocation.

16. Pseudocode

```
START

Create an initial allocation of vehicles to parking spaces
Calculate H(current_allocation)

REPEAT
    Generate a neighboring allocation
    Calculate H(neighbor_allocation)

    IF H(neighbor_allocation) < H(current_allocation) THEN
        Accept neighbor_allocation as current_allocation
    ELSE
        Keep current_allocation
    END IF
```

```
UNTIL no better neighboring allocation exists

Display current_allocation

STOP
```

17. Working of the Method

- The system first selects an available allocation as the starting solution.
- It calculates the allocation's heuristic score.
- A nearby alternative allocation is generated by changing one parking space.
- The alternative allocation is evaluated using the same heuristic function.
- If the new allocation has a lower score, the system moves to that allocation.
- The process continues until an improvement cannot be found.

18. Limitation of Hill Climbing

Hill Climbing can stop at a local optimum. This means that it may find an allocation that cannot be improved by moving a single vehicle even though another arrangement of several vehicles could have a lower overall score. Therefore, the quality of the final result depends on the starting allocation and the neighboring allocations considered.

19. Advantages

- The heuristic function is simple and easy to calculate.
- Multiple factors can be considered at the same time.
- The method can reduce time spent searching for a parking space.
- The weights can be changed according to campus requirements.
- The approach can be adapted to different parking layouts.
- The method can support a basic allocation system without requiring every possible allocation to be examined.

20. Limitations

- The result depends on the quality of the selected weights.
- Congestion levels change during the day.
- A space may be taken by another vehicle before the driver reaches it.
- Hill Climbing may reach a local optimum.

- The system needs updated information when spaces become occupied or free.

21. Possible Improvements

- Use live occupancy information from sensors or entry gates.
- Add accessibility information such as spaces for differently-abled drivers.
- Add event-day rules and temporary parking restrictions.
- Include electric vehicle charging spaces for vehicles that need them.
- Use a more advanced search strategy when a larger campus is involved.

22. Conclusion

A heuristic function provides a practical way to solve the parking space allocation problem. A campus can have many parking spaces that a vehicle could use, and selecting a space only because it is empty or nearest may not give the most suitable result. By combining walking distance, size mismatch, congestion and restrictions, the proposed heuristic provides a single score that can be used to compare allocations.

In this assignment, a lower heuristic value represents a more suitable allocation. The example demonstrates how two parking spaces can be evaluated using the same formula. A Hill Climbing approach can then use this heuristic to repeatedly improve an allocation until no better neighboring allocation is found.

The proposed approach is simple enough for a basic college parking system and can be extended when additional information becomes available. Overall, heuristic-based allocation demonstrates how Artificial Intelligence can be applied to a practical problem in a college environment.

23. Summary

The assignment developed a heuristic function for parking space allocation, defined the required inputs and objectives, demonstrated the calculation using two parking spaces, and described a Hill Climbing method and pseudocode for improving the selected allocation.